Automated Irregular Shaped Object Stacking via Pose Optimization and Monte Carlo Tree Search

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***Abstract*—We study automated stacking of irregular objects to approximate prescribed target shapes without support from container walls. Our framework combines bilevel pose optimization with batched, receding-horizon Monte Carlo Tree Search (MCTS) over object choices and placement poses. The optimizer couples geometric objectives with contact-force optimization to evaluate force and torque equilibrium across the structure under friction constraints, using a bodywise decomposition for GPU parallelization. Physical simulation validates placements and supplies settled successor states for multi-step search. We hypothesize that combining structural equilibrium evaluation with lookahead improves target coverage while maintaining stability under a fixed planning budget.**

I. Introduction

Automated packing and palletizing have long been studied to improve space utilization and material handling in logistics and manufacturing [6]. Recent robotic packing methods accommodate irregular objects while considering packing density, stability, and placement feasibility [22, 9]. Constructing freestanding structures from natural stones requires finding mutually supporting arrangements among uneven surfaces while achieving a prescribed target shape without support from container walls.

Several stone-stacking methods use placement criteria developed for particular construction settings, including tower-building objectives [7] and masonry-inspired geometric rules [12, 19]. For example, Johns et al.'s double-faced wall planner generates candidates along the intersection of the target wall surfaces and the existing structure, and applies inward-slope heuristics tailored to wall construction [12]. The ability to construct freeform walls [13] demonstrates flexibility in wall geometry, while extending beyond such construction settings motivates planning driven by target geometry without wall-specific placement rules. Structural stability also requires reasoning beyond the support of the newly placed object: its load can affect the equilibrium of objects beneath it. Liu and Napp [17] explicitly address structural strength during planning, demonstrating the value of evaluating the structure as a whole. However, assessing the current structure alone does not determine how a placement affects future support and space utilization. Menezes et al. [20] address sequential placement through learned action values that account for future rewards, with a prescribed object sequence and fixed orientations. We investigate explicit lookahead over object choices and placement poses, coupled with structural equilibrium evaluation and target-shape objectives.

We consider stacking a finite inventory of n irregular objects to approximate a target shape $T \subset \mathbb{R}^3$. Let $O_i(q_i) \subset \mathbb{R}^3$ denote the volume occupied by object i at pose $q_i \in \mathrm{SE}(3)$. The final configuration objective is

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathbb{V}\left(T \setminus \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \tag{1}$$
$$c_{\text{phys}}(O_1(q_1),\cdots,O_n(q_n)) = 0,$$

where $\mathbb{V}(\cdot)$ denotes volume and $c_{\text{phys}} = 0$ represents stability and non-interpenetration constraints. Jointly optimizing all object poses under coupled physical constraints is computationally demanding. To obtain a practical planning procedure, we decompose construction into sequential decisions, selecting one object and optimizing its placement at each step while checking the resulting structure's physical feasibility. This decomposition reduces the dimension of each pose optimization, but each placement changes the support geometry, free space, and remaining inventory. Multi-step lookahead allows the planner to evaluate these future consequences beyond immediate placement quality. Our central question is how to combine this lookahead with structural equilibrium evaluation in a common framework for different target shapes.

We address this question with a framework coupling bilevel pose optimization and Monte Carlo Tree Search (MCTS). The pose optimizer refines geometric proposals using target-shape objectives and a physical objective that evaluates force and torque equilibrium across the structure under friction constraints. The target enters through its geometric representation, allowing the same optimization formulation to be applied to different target shapes. A bodywise ADMM decomposition enables parallel solution of the contact-force subproblem.



Fig. 1: Preliminary stacking results. Left: simulated structures for different target shapes. Right: a physical stone structure from an outdoor experiment.

The sequence planner embeds pose refinement and physical simulation in batched, receding-horizon MCTS, following prior work on batched simulation-based search [8]. Simulation supplies settled successor states and rejects unstable placements, while multi-step lookahead compares their effects on subsequent construction.

We hypothesize that combining structural equilibrium evaluation with multi-step lookahead improves target coverage while maintaining stability across different target geometries under a fixed planning budget. Comparisons with greedy placement and candidate-only equilibrium evaluation will test the effects of lookahead and structural coupling, respectively. Evaluations across target shapes and search horizons will examine the applicability of the common formulation and its computational tradeoffs. The proposed contributions are:

- A bilevel placement optimizer that combines explicit target geometry with contact-force optimization across the structure, using bodywise ADMM updates for GPU computation.
- A batched, receding-horizon stacking planner that integrates pose refinement and physical simulation with multi-step lookahead over object choices and placement poses.

II. Related Works

A. Automated Stone Stacking

Furrer et al. [7] combined gradient-based pose search with physics simulation for stone stacking. Johns et al. [12] integrated perception, placement planning, and excavator manipulation for wall construction, later combining constrained registration and signed-distance-field classification to plan prescribed wall geometries [13]. Liu et al. [19] selected simulation-generated poses using hierarchical filters informed by masonry heuristics. Liu and Napp [17] incorporated structural strength evaluation into sequence planning and validated simulated predictions through physical wall tests. Their subsequent work introduced stackability measures for stone selection in corbelled shelter construction [18]. Menezes et al. [20] learned placement value maps from elevation maps using deep Q-learning, with rewards accounting for target occupancy and object displacement. Their policy selects planar positions for a prescribed object sequence with fixed orientations; its observed construction strategies inform our initial pose sampler.

B. Irregular Object Packing

Wang and Hauser [22] used heightmap minimization to pack nonconvex objects while considering stability and robot motion feasibility, with conservative planning and replanning to address execution errors. Huang et al. [9] jointly planned object order and placement through hierarchical reinforcement learning using multi-view heightmaps and rewards for compactness, pyramidality, and simulated stability. Such geometric guidance is relevant to generating initial placement candidates before continuous pose refinement.

C. Simulation-Based Sequential Manipulation Planning

King et al. [14] extended MCTS to nonprehensile rearrangement under uncertainty using learned and heuristic rollout policies. Huang et al. [8] introduced Parallel Monte Carlo Tree Search with Batched Simulations (PMBS), coupling parallel search with GPU rigid-body simulation for object retrieval from clutter. Our stacking planner adopts batched evaluation, combining geometric sampling, equilibrium-aware pose refinement, and structural simulation at each expansion.

III. Overview and Preliminaries

A. Shape Representation

We represent the target T and each object O_i as unions of smooth convex bodies modeled by differentiable support functions (DSFs) [15]. For a compact convex body $C \subset \mathbb{R}^3$, its support function is $h_C(x) = \max_{p \in C} x^\top p$ for $x \neq 0$. At pose $q = (R, t) \in \mathrm{SE}(3)$, we write

$$C(q) = \{Rp + t \mid p \in C\}, \qquad O_i(q_i) = \bigcup_j C_{ij}(q_i),$$

where C_{ij} are the convex components in object i's local frame. Similarly, $T = \bigcup_j T_j$, where T_j are fixed convex components in the world frame. DSFs support differentiable signed-distance and contact-geometry computations between posed convex bodies [2], enabling gradient-based pose optimization.

B. Markov Decision Process Formulation

To address the construction objective in Eq. (1), we model stacking as a finite-horizon sequential decision process in which one object is placed at each step. Let $\mathcal{X}_k = \{(i, q_i) \mid i \in \mathcal{P}_k\}$ be the current scene and $\mathcal{I}_k = \{1, \dots, n\} \setminus \mathcal{P}_k$ the remaining inventory. The state, action, and deterministic transition function are

$$s_k = (\mathcal{X}_k, \mathcal{I}_k, T), \quad a_k = (i_{k+1}, q_{k+1}), \quad s_{k+1} = F(s_k, a_k),$$

where $i_{k+1} \in \mathcal{I}_k$. The transition inserts the proposed pose into the scene, simulates the resulting structure, and returns the settled scene with $\mathcal{I}_{k+1} = \mathcal{I}_k \setminus \{i_{k+1}\}$. The reward $r(s_k, a_k)$ measures construction progress and physical validity; its implementation is described in Sec. V. The sequence planner therefore operates on the MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, F, r, \gamma)$.

C. Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) incrementally constructs a search tree $\mathcal{T} = (\mathcal{N}, \mathcal{E})$ over the MDP. Each node $n \in \mathcal{N}$ represents a state $s(n)$, and each edge $(n, a) \in \mathcal{E}$ represents an action and its transition to a child node. Let $N(n)$ denote the number of completed evaluations and $Q(n)$ the empirical mean of discounted returns backpropagated to that node.

Each iteration comprises four operations. During *selection*, the tree policy follows promising children by balancing empirical value and exploration. During *expansion*, a new action is generated at the selected node; *evaluation* obtains its reward and successor state. A nonterminal leaf may also receive a value estimate $\hat{V}(s)$ from a rollout or problem-specific

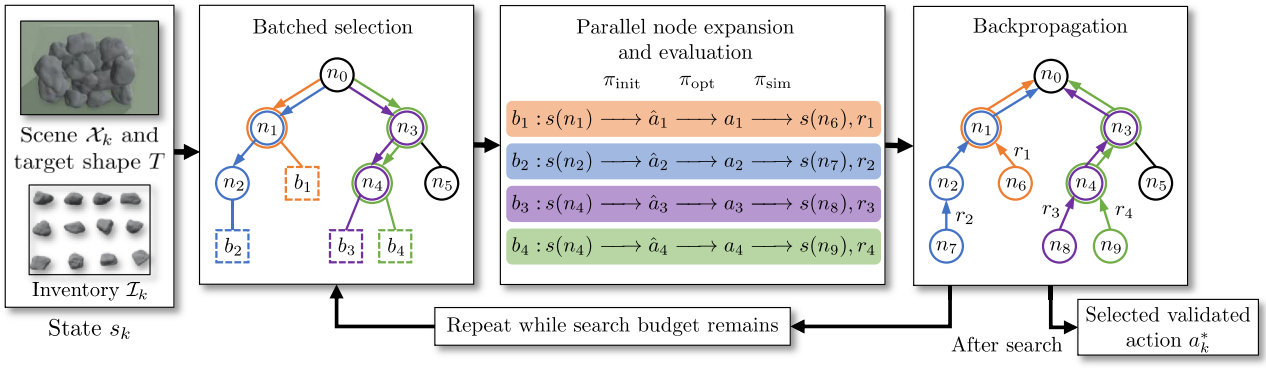


Fig. 2: Batched MCTS sequence planning. Colors identify expansion requests b_j and their tree paths. Requests are evaluated in parallel through π_{init}, π_{opt}, and π_{sim}, and returns are backpropagated along the selected paths. Search repeats within the budget before selecting a physically validated root action a_k^*.

heuristic. Finally, *backpropagation* propagates the discounted return along the selected path to update $Q(n)$ and $N(n)$.

Here, actions place remaining objects, with evaluation involving pose refinement and physical simulation. Our geometry-informed proposals, batched selection, and finite-horizon leaf evaluation are described in Sec. V.

D. Overall Structure of the Sequence Planner

Figure 2 illustrates the MCTS sequence planner. For each expansion, the initial pose sampler π_{init} proposes a candidate action $\hat{a}_k = (i_{k+1}, \hat{q}_{k+1})$, specifying both the object index and its initial pose. The pose optimization module π_{opt} refines the pose to obtain $a_k = (i_{k+1}, q_{k+1})$. The structural simulation module π_{sim} evaluates the entire scene and produces the settled successor state. Validity and simulation statistics determine tree admission and reward, giving the execution flow

$$s_k \xrightarrow{\pi_{\text{init}}} \hat{a}_k \xrightarrow{\pi_{\text{opt}}} a_k \xrightarrow{\pi_{\text{sim}}} s_{k+1},\ r(s_k, a_k).$$

We describe pose optimization in Sec. IV and sequence planning, including structural simulation and reward design, in Sec. V.

IV. Pose Optimization

The module π_{opt} refines $\hat{q}_{k+1}$ from π_{init} toward a stable, physically feasible placement q_{k+1}.

A. Single Object Pose Optimization Problem

Given s_k, we initialize q at $\hat{q}_{k+1}$ and solve the bilevel problem

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \le 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

where $q \in \text{SE}(3)$ is the candidate object's pose and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ collects all contact forces in the resulting structure, including inter-object and ground contacts. This formulation follows prior contact-aware optimization methods [16, 5].

The geometric objective f_{geo} is designed as follows:

$$\begin{aligned} f_{\text{geo}}(q) = w_{\text{h}} f_{\text{h}}(q) + w_{\text{in}} f_{\text{in}}(q \mid T, O_{i_{k+1}}) \\ + w_{\text{pen}} f_{\text{pen}}(q \mid \mathcal{X}_k, O_{i_{k+1}}) \end{aligned}$$

with positive weights. Here, f_{h} penalizes higher placements and f_{pen} penalizes penetration into the current scene. Writing $O = O_{i_{k+1}}$ and $h_{O(q)}(u) = \max_j h_{C_{i_{k+1},j}(q)}(u)$, we define

$$\begin{aligned} m_{\text{in}}(q) &= \max_j \min_{u \in \mathbb{S}^2} \left[h_{T_j}(u) - h_{O(q)}(u) \right], \\ f_{\text{in}}(q) &= \left[\min\{ m_{\text{in}}(q), 0 \} \right]^2. \end{aligned}$$

This term penalizes violations of containment of the entire object within a single convex target component, a conservative condition for containment in T. It is differentiable almost everywhere in q; we compute pose derivatives at regular inner solutions by implicit differentiation. Nonsmooth transitions have not caused practical difficulties in our implementation.

Let $\mathcal{P}_{k+1} := \mathcal{P}_k \cup \{i_{k+1}\}$ and let $i_0 \notin \mathcal{P}_{k+1}$ index the fixed ground. For distinct $l, m \in \mathcal{P}_{k+1} \cup \{i_0\}$, let $I_{l,m} \subset \{1, \ldots, n_c\}$ index contacts between bodies l and m. Each object's contact set $I_l := \bigcup_{m \in (\mathcal{P}_{k+1} \cup \{i_0\}) \setminus \{l\}} I_{l,m}$ includes ground contacts, but the physical objective excludes the ground's equilibrium residual:

$$\begin{aligned} f_{\text{phys}}(q, \lambda) = w_{\text{eq}} \sum_{l \in \mathcal{P}_{k+1}} \| J_{c,I_l}^{\top}(q) \lambda_{I_l} + \lambda_{\text{ext},l} \|^2 \\ + w_{\text{cp}} \sum_{j \in I_{i_{k+1}}} \mathrm{d}_j(q)^2 \|\lambda_j\|^2 + \rho \sum_{j=1}^{n_c} \|\lambda_j\|^2 \end{aligned} \tag{3}$$

The three terms respectively penalize violations of static equilibrium, contact complementarity between the candidate and the current scene, and large contact forces through regularization. Here, $\mathrm{d}_j(q)$ is the signed distance between the two bodies associated with contact candidate j. J_{c,I_l} is the contact Jacobian for object O_l, $\lambda_{\text{ext},l}$ is its external wrench, λ_{I_l} is the vector of contact forces at the contact points in I_l, and $w_{\text{eq}}, w_{\text{cp}}, \rho > 0$ are weighting and regularization parameters.

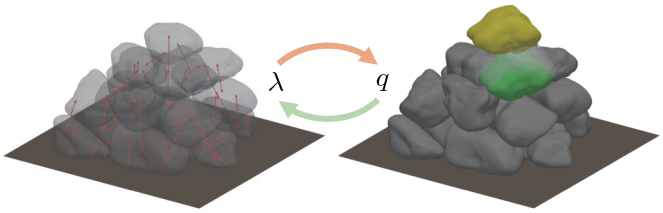


Fig. 3: Bilevel pose optimization. (Left) For a fixed candidate pose q, the lower-level problem optimizes contact forces λ under friction constraints to minimize structural equilibrium residuals. (Right) The upper-level problem refines $q \in \mathrm{SE}(3)$ using the geometric objective and the optimized physical objective.

Finally, each $g_{\text{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\text{cone},j}(q, \lambda_j) = \sqrt{\|J_{c,j}^t \lambda_j\|^2 + \epsilon} - \mu_j J_{c,j}^n \lambda_j \tag{4}$$

where $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J_{c,j}^t$ and $J_{c,j}^n$ are the tangential and normal components of the Jacobian of the contact constraint.

B. Upper-Level Problem Optimization on SE(3) Manifold

Denoting $\lambda^*(q)$ as the optimal λ of the lower-level problem for a given pose q, we define the force dependent objective function as $f_{\text{phys}}(q) := f_{\text{phys}}(q, \lambda^*(q))$. Then, using the *Envelope theorem* [4], we can compute the gradient of $f_{\text{phys}}(q)$ with respect to q as follows:

$$\frac{df_{\text{phys}}(q)}{dq} = \frac{\partial f_{\text{phys}}(q, \lambda^*)}{\partial q} + \sum_{j=1}^{n_c} \nu_j^* \frac{\partial g_{\text{cone},j}(q, \lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable associated with the j-th friction cone constraint $g_{\text{cone},j}(q, \lambda_j^*) \leq 0$. Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with BFGS Hessian approximation [10]. In addition, motivated by homotopy based optimization methods [], we designed the sequence of the weight parameters in $f_{\text{geo}}(q)$ and $f_{\text{phys}}(q)$ such as w_{h}, w_{in}, w_{pen}, w_{eq}, w_{cp} and iteratively solve the bi-level optimization problem (2) to induce a better local minimum.

C. Lower-Level Problem Optimization with ADMM

For a fixed q, the lower-level problem is a convex optimization problem with respect to λ which can be efficiently solved. Moreover, exploiting the inherent structure of the lower-level problem, we can reformulate it to be separable with respect to the objects $\mathcal{P}_{k+1}$ in the scene and apply the alternating direction method of multipliers (ADMM) [3] to solve it in a scalable manner. For each contact force λ_j, we introduce two local copies $\lambda_{j,1}$ and $\lambda_{j,2}$ associated with the two contacting bodies, including the fixed ground. Specifically, for $j \in I_{l,m}$ with $l < m$, the copies are associated with bodies l and m, respectively. We then impose the consensus constraints $\lambda_{j,1} = \lambda_{j,2}$. Collecting all local copies into $\tilde{\lambda}$ makes the force

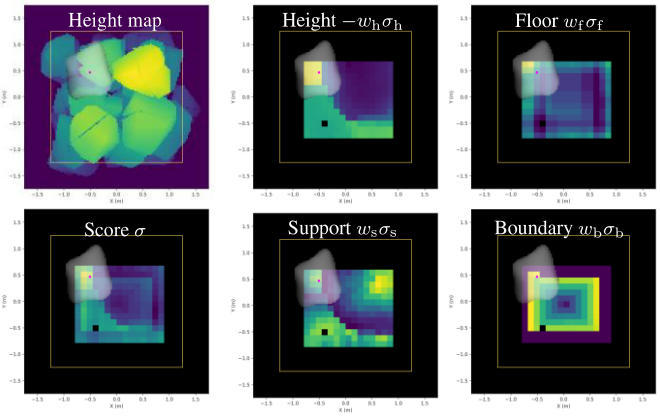


Fig. 4: Initial pose sampling for a fixed object and orientation. Left: scene height (top) and proposal score (bottom). Right: weighted height, floor, support, and boundary contributions.

and torque equilibrium term separable by body: each body's residual depends only on its own local contact-force variables. The consensus constraints maintain agreement between copies of each shared contact force. The lower-level problem can therefore be reformulated as

$$\begin{aligned} \min_{\tilde{\lambda}, z} \quad & \tilde{f}_{\text{phys}}(q, \tilde{\lambda}) + \sum_{\substack{j=1,\dots,n_c \\ s=1,2}} \delta_{K_j}(z_{j,s}) \\ \text{s.t.} \quad & \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2}, \qquad j = 1, \dots, n_c. \end{aligned}$$

where $\tilde{f}_{\text{phys}}(q, \tilde{\lambda})$ is the extended physical force function and δ_{K_j} is the indicator function of the convex set of feasible contact forces $K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\text{cone},j}(q, \lambda) \leq 0\}$. Here, $z_{j,s}$ is an auxiliary variable associated with the j-th contact point and $s \in \{1, 2\}$. At each ADMM iteration, the objectwise-separable subproblem decomposes into $|\mathcal{P}_{k+1}|$ independent subproblems, one for each object O_l. An equivalent four-dimensional second-order cone (SOC) lifting enables closed-form cone projections for the friction updates, while consensus updates reduce to averaging.

The pose optimization module is implemented using CUDA to leverage GPU parallelization for efficient computation of the gradients and the ADMM updates.

V. Sequence Planning

In this section, we describe initial pose sampling, structural simulation, and reward design, followed by their integration into batched MCTS.

A. Initial Pose Sampling

The sampler proposes an object i, orientation R, and planar location $\mathbf{x}$ on a feasible grid $\mathcal{G}_{i,R}(s_k)$ according to

$$\begin{aligned} &\pi_{\text{init}}(i, R, \mathbf{x} \mid s_k) = \pi_{xy}(\mathbf{x} \mid i, R, s_k) \pi_{\text{unif}}(i, R \mid s_k), \\ &\pi_{xy}(\mathbf{x} \mid i, R, s_k) \propto \exp(\sigma(\mathbf{x} \mid i, R, s_k)/\tau)\,. \end{aligned}$$

where π_{unif} is uniform over the remaining objects $i \in \mathcal{I}_k$ and finite rotation set $\mathcal{R}$, and $\tau > 0$ controls the preference for high scores. Scene geometry determines the initial height $z_{i,R}$,

yielding $\hat{q}_{k+1} = (R, (\mathbf{x}, z_{i,R}(\mathbf{x}; s_k)))$ for refinement by π_{opt}. Omitting the state dependence, the proposal score is

$$\sigma(\mathbf{x} \mid i_{k+1}, R) = w_{\text{s}}\sigma_{\text{s}}(\mathbf{x} \mid i_{k+1}, R) + w_{\text{f}}\sigma_{\text{f}}(\mathbf{x} \mid i_{k+1}, R) + w_{\text{b}}\sigma_{\text{b}}(\mathbf{x} \mid i_{k+1}, R) - w_{\text{h}}\sigma_{\text{h}}(\mathbf{x} \mid i_{k+1}, R),$$

with positive weights. The support score σ_{s} favors larger support areas, σ_{f} rewards filling empty floor area, σ_{b} favors the target boundary, and σ_{h} penalizes height. The floor and boundary terms are inspired by the dense first-course packing and boundary-first behavior observed by Menezes et al. [20]; the height penalty follows the low-placement preference of their geometric baseline. We use these preferences to sample initial poses for continuous refinement. Figure 4 illustrates these score maps; their computation is deferred to the appendix.

B. Simulation Module

After π_{opt} refines the candidate pose $\hat{q}_{k+1}$, the simulation module π_{sim} evaluates the deterministic transition $s_{k+1} = F(s_k, a_k)$. It first simulates the new object with the existing scene fixed to refine the placement, and then simulates the entire scene to verify global stability.

Both stages use early termination. The simulation stops when all placed objects have sufficiently small kinetic energy or when accumulated motion energy indicates instability. For object O_i, we define

$$E_{\text{kin},i}(t) = \frac{1}{2m_i}\mathcal{V}_i^\top M_i \mathcal{V}_i, \quad E_{\text{acc},i}(t) = \sum_{s=0}^{t} \sqrt{E_{\text{kin},i}(s)}\Delta t$$

where $\mathcal{V}_i$ is the spatial velocity, M_i and m_i are the mass matrix and mass, respectively, and Δt is the simulation time step. The early-termination condition is

$$\max_{i \in \mathcal{P}_{k+1}} E_{\text{kin},i}(t) < \epsilon_{\text{kin}} \quad \text{or} \quad \max_{i \in \mathcal{P}_{k+1}} E_{\text{acc},i}(t) > \epsilon_{\text{acc}} \tag{5}$$

with the thresholds $\epsilon_{\text{kin}}, \epsilon_{\text{acc}} > 0$. We implement π_{sim} in CUDA using the GPU-accelerated subADMM algorithm [11].

C. Reward Design

The reward balances target coverage, object containment, surface uniformity, and physical stability:

$$r(s_k, a_k) = w_T \frac{\mathbb{V}\left(\overline{\mathcal{X}}_{k+1}\right) - \mathbb{V}\left(\overline{\mathcal{X}}_k\right)}{\mathbb{V}(T)} + w_O \frac{\mathbb{V}\left(O_{i_{k+1}} \cap T\right)}{\mathbb{V}\left(O_{i_{k+1}}\right)} - w_H \left(\max_{\mathbf{x}} H_{\mathcal{X}_{k+1}}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}[H_{\mathcal{X}_{k+1}}(\mathbf{x})]\right) - w_D \mathbb{E}_{\mathcal{V} \sim \mathcal{N}(0,\Sigma)}\left[\Delta q_{k+1}(\mathcal{V})\right] - w_S f_{\text{eq}}$$

where all weights are positive, $\overline{\mathcal{X}}_k := T \cap \bigcup_{i \in \mathcal{P}_k} O_i(q_i)$ is the occupied portion of the target shape, and $H_{\mathcal{X}_{k+1}}(\mathbf{x})$ is the upper surface height at $\mathbf{x} = (x, y)$. The first two terms reward newly filled target volume and the fraction of the candidate contained in the target, respectively. The remaining terms penalize surface height variation, the candidate's expected pose deviation $\Delta q_{k+1}(\mathcal{V})$ under Gaussian perturbations $\mathcal{V} \sim \mathcal{N}(0, \Sigma)$, and the equilibrium residual f_{eq} from π_{opt}.

Algorithm 1 Batched receding-horizon MCTS for sequence planning

Require: State s_k, previous search tree $\mathcal{T}$, batch size B
Ensure: Next state s_{k+1} and updated tree $\mathcal{T}$

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Initialize or update T at s_k and evaluate root candidates
while the search budget remains do
    B ← SELECTBATCH(T, B)
    E ← EVALUATEBATCH(B, π_init, π_opt, π_sim)
    T ← BACKPROPAGATE(T, B, E)
end while
a* ← arg max_{a ∈ A_val(s_k)} Q(n_a)
Execute a* and obtain s_{k+1}
Promote its child to the root and retain compatible descendants
return s_{k+1}, T
```

D. Monte Carlo Tree Search

We combine π_{init}, π_{opt}, and π_{sim} in a batched, receding-horizon MCTS planner over the MDP defined in Sec. III. To amortize expensive pose optimization and simulation, we evaluate candidate placements in parallel, following the batched planning approach of [8]. Algorithm 1 summarizes the finite-horizon search and execution procedure, starting from a tree rooted at the current state s_k.

Batched Selection. We select a batch of expansion paths using PUCT [21] with geometry- and equilibrium-informed action priors. For a child n reached from n_p by action a, selection maximizes

$$\text{PUCT}(n) = \frac{N(n)Q(n)}{N(n) + \hat{N}(n)} + cP(n_p, a)\frac{\sqrt{N(n_p) + \hat{N}(n_p)}}{1 + N(n) + \hat{N}(n)}, \tag{6}$$

where Q is the mean return, N and $\hat{N}$ are completed and temporary virtual visit counts, $P(n_p, a)$ is the action prior at the parent (distinct from the transition function F), and $c > 0$ controls exploration. Progressive widening allows more children as completed visits increase, while virtual visits diversify selection within each batch. All paths are selected before physical evaluation begins.

Parallel Expansion and Evaluation. Each selected expansion applies π_{init}, π_{opt}, and π_{sim} to generate, refine, and physically evaluate a candidate placement. These evaluations run in parallel and provide settled successor states, validity outcomes, and rewards for tree expansion and backpropagation. Root candidates are evaluated through the same pipeline.

Leaf Value Estimation and Backpropagation. To avoid further optimization and simulation costs, we replace additional rollouts with a heuristic estimate of the remaining return. For a nonterminal leaf s_l at depth d_l from the current root, we use

$$\hat{V}(s_l) = \bar{r}\sum_{j=0}^{h_l - 1} \gamma^j, \qquad h_l = \min\{d_M - d_l, |\mathcal{I}(s_l)|\},$$

where d_M is the maximum search depth, $\mathcal{I}(s_l)$ is the remaining inventory, and $\bar{r}$ is the mean reward estimated from

evaluated placements. Terminal and failed leaves receive no bootstrap value. Discounted placement rewards and the leaf estimate are backpropagated along each selected path to update mean returns Q and visit counts N. Temporary virtual visits are then removed.

Action Selection and Tree Reuse. After the search budget is exhausted, we execute the physically validated root action with the highest mean return. In Algorithm 1, $\mathcal{A}_{\text{val}}(s_k)$ denotes these validated actions and n_a the child reached by a. The selected child becomes the new root, retaining descendants and search statistics compatible with the resulting state. Planning then repeats with a finite lookahead horizon.

Acknowledgments

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